
DARAXONRASIB EFFICIENT LLM TRIAL SIMULATIONS

Kevin Kawchak 

Chief Executive Officer

ChemicalQDevice

San Diego, CA

Dec 30, 2025

kevink@chemicalqdevice.com

*FDA: Metastatic Pancreatic Cancer Drug Daraxonrasib, Phase 1 Breakthrough Therapy
Nature Medicine: Revolution Medicines Phase 3 Daraxonrasib - Top 2026 Clinical Trial
LLMs: Identified Daraxonrasib & Simulated 3 Multi-Arm PDAC Trials With Validations*

FDA 06/25 [1], Nature 12/25 [2]. LLMs: Daraxonrasib ID 06/25; Simulations & Validations 07/25, 08/25, 10/25 [3, 4, 5, 6]

Daraxonrasib Identification [3]

- 1) ChatGPT o3 Deep Research meta-analyses combine for >400,000 words for 40 Top PDAC clinical trial literature areas
- 2) Gemini 2.5 Pro utilized meta-analyses to predict top untested Daraxonrasib + Mitazalimab + nal-IRI triplet combination
- 3) Gemini Therapeutic Synergy & Viability Score (TSVS) = 8.15 based on Efficacy, Toxicity, MoA, and Biomarker metrics

Daraxonrasib Simulations [4, 5, 6]

- 1) ChatGPT 100,000 Patient 24-Month In Silico Phase III 5-Arm Pancreatic Cancer Clinical Trial Triplicate
 - Advantages: Large and diverse patient population, FOLFIRINOX [7] and nal-IRI [8] controls, Code-less LLM prompting
 - Results: G3+ AE: Daraxonrasib doublet (8.0%) over FOLFIRINOX (25.0%) in PRODIGE-4. FOLFIRINOX best HR = 0.57
 - Limitation: KRAS G12C log file (favors exp) vs. KRAS-mutant report (favors control) discrepancy. Trial to trial variability
- 2) QSP Metastatic Pancreatic Cancer AI Clinical Trial Simulation From Protocol to Prediction: Code, VVUQ, and Playbook
 - Advantages: Biologically complex QSP, 250 ODEs/patient, 10-arm, patient diversity. nal-IRI [8], nab-Paclitaxel [9] controls
 - Results: Reproducible, Daraxonrasib triplet best mOS HR 12.8 (0.25), doublet 12.8 (0.23). 6/7 patient archetypes OS HR <1
 - Limitations: Assumed no acquired resistance and ideal pharmacodynamics. G3+ AE consistently high for all patient arms
- 3) Accelerating FDA Compliance & Cost Efficiency of in silico Clinical Trials via AI Digital Twin Pancreatic Cancer Simulation
 - Advantages: Digital twin bidirectional 1000 patient KRAS-G12C/KRAS-G12D code; nal-IRI and nab-Paclitaxel controls
 - Results: Daraxonrasib triplet best mPFS HR 7.5 (0.31), high mOS HR 12.1 (0.34) prior to maintenance therapy transition
 - Limitations: No patient-specific PK/PD or tumor growth parameters; Simple Emax PK/PD, no direct immune compartments

Daraxonrasib Validations [4, 5, 6]

- 1) ChatGPT 100,000 Patient 24-Month In Silico Phase III 5-Arm Pancreatic Cancer Clinical Trial Triplicate
 - ChatGPT o3 pro Deep Research: ICH E3-aligned trial reports, log files, and Flatiron Health external validations [10]
 - Gemini, Grok, OpenAI & Claude cross-verifications. 5 arm consistency = 8.95-9.45. Overall trial reproducibility = 8.65
- 2) QSP Metastatic Pancreatic Cancer AI Clinical Trial Simulation From Protocol to Prediction: Code, VVUQ, and Playbook
 - Numerically stable objective response rates observed for final trial time step $dt = 0.05$ and tumor grid size = 5
 - Opus 4.1: Sensitivity analysis of targeted therapy vs. control, similar trends for increasing resistance factor from 5% to 30%
- 3) Accelerating FDA Compliance & Cost Efficiency of in silico Clinical Trials via AI Digital Twin Pancreatic Cancer Simulation
 - FDA M15 and ASME V&V 40 aligned verification, validation, uncertainty quantification. Overall credibility score = 81.9
 - Gemini 2.5 Pro: 55 VVUQ Python notebook tests address RECIST, Seed, Drug Params, Regimen Toxicity, Archetypes

LLM Financial Assessments [4, 5, 6]

- Industry: Single run virtual trial = \$120,000+, 4.5 months. Author: Triple empirical virtual trials \$36,330 (est.), 1 month
- Industry: QSP virtual trial = \$2M+, several months to 1+ year. Author: 10-arm QSP trial \$36,330 (est.), 1 month
- Industry: DT Simulator 3-16 wks, \$28K-\$700K (lean-prototype) [11]. Author: Phase II platform \$36,330 (est.), 1 month

Scaling LLM Simulations [12, 13, 14]

- Utilize Jan 26 "Scaling Oncology LLM Generated Code" guidance to generate quality 1000+ line Python trial simulations
- Implement "End-to-End Oncology Clinical Trial LLM Efficiency" recommendations for next generation LLM applications
- Utilize AI peer review throughout work for rapid progress. Maximize efficiency by minimizing humans, prompts, cost, time

Cite This Work

- [1] Revolution Medicines. Revolution medicines announces fda breakthrough therapy designation for daraxonrasib in previously treated metastatic pancreatic cancer with kras g12 mutations, June 2025. URL: <https://ir.revmed.com/news-releases/news-release-details/revolution-medicines-announces-fda-breakthrough-therapy>.
- [2] Mike May. Eleven clinical trials that will shape medicine in 2026. *Nature Medicine*, 31(12):3943–3947, December 2025. URL: <https://www.nature.com/articles/s41591-025-04083-x>, doi:10.1038/s41591-025-04083-x.
- [3] Kevin Kawchak. End-to-end pancreatic ductal adenocarcinoma digital twin clinical trial proposals, June 2025. doi:10.5281/zenodo.15735068.
- [4] Kevin Kawchak. Chatgpt 100,000 patient 24-month in silico phase iii 5-arm pancreatic cancer clinical trial triplicate, July 2025. doi:10.5281/zenodo.16415815.
- [5] Kevin Kawchak. Qsp metastatic pancreatic cancer ai clinical trial simulation from protocol to prediction: Code, vvuq, and playbook, August 2025. doi:10.5281/zenodo.17001137.
- [6] Kevin Kawchak. Accelerating fda compliance and cost efficiency of in silico clinical trials via ai digital twin pancreatic cancer simulation, September 2025. doi:10.5281/zenodo.17239510.
- [7] Sophie Gourgou-Bourgade, Caroline Bascoul-Mollevi, Françoise Desseigne, Marc Ychou, Olivier Bouché, Rosine Guimbaud, Yves Bécouarn, Antoine Adenis, Jean-Luc Raoul, Valérie Boige, Jocelyne Bérille, and Thierry Conroy. Impact of folfirinix compared with gemcitabine on quality of life in patients with metastatic pancreatic cancer: Results from the prodige 4/accord 11 randomized trial. *Journal of Clinical Oncology*, 31(1):23–29, January 2013. URL: <https://ascopubs.org/doi/10.1200/JCO.2012.44.4869>, doi:10.1200/JCO.2012.44.4869.
- [8] Andrea Wang-Gillam, Richard A. Hubner, Jens T. Siveke, Daniel D. Von Hoff, Bruce Belanger, Floris A. de Jong, Beloo Mirakhur, and Li-Tzong Chen. Napoli-1 phase 3 study of liposomal irinotecan in metastatic pancreatic cancer: Final overall survival analysis and characteristics of long-term survivors. *European Journal of Cancer*, 108:78–87, February 2019. URL: <https://www.sciencedirect.com/science/article/pii/S0959804918315533>, doi:10.1016/j.ejca.2018.12.007.
- [9] Manuel Hidalgo, Carlos Plaza, Monica Musteanu, Peter Illei, Carrie B. Brachmann, Carla Heise, Daniel Pierce, Pedro P. Lopez-Casas, Camino Menendez, Josep Tabernero, Alfredo Romano, Xinyu Wei, Fernando Lopez-Rios, and Daniel D. Von Hoff. Sparc expression did not predict efficacy of nab -paclitaxel plus gemcitabine or gemcitabine alone for metastatic pancreatic cancer in an exploratory analysis of the phase iii impact trial. *Clinical Cancer Research*, 21(21):4811–4818, November 2015. URL: <https://aacrjournals.org/clincancerres/article/21/21/4811/175077/SPARC-Expression-Did-Not-Predict-Efficacy-of-nab>, doi:10.1158/1078-0432.CCR-14-3222.
- [10] Flatiron Health. <https://meetinglibrary.asco.org/record/165000/abstract> authors: Sources: Asco annual meeting. URL: <https://tinyurl.com/582czksc>.
- [11] OpenAI. Introducing gpt-5.2. the most advanced frontier model for professional work and long-running agents., December 2025. URL: <https://openai.com/index/introducing-gpt-5-2/>.
- [12] Kevin Kawchak. Scaling oncology llm generated code: Jan 2026, December 2025. doi:10.5281/zenodo.18072433.
- [13] Kevin Kawchak. End-to-end oncology clinical trial llm efficiency for industry adoption with fda/ich regulations, October 2025. doi:10.5281/zenodo.17451709.
- [14] Kevin Kawchak. Ai peer review acceleration of llm-generated glioblastoma clinical trial patient matching ml, fda/ich/iso, and fastapi, November 2025. doi:10.5281/zenodo.17774560.

